

Experiment 3: Design Zero-shot, One-shot, and Few-shot Prompts to Summarize an Article into 50 Words

Aim

To implement Zero-shot, One-shot, and Few-shot prompting techniques using the OpenAI API to summarize an article into 50 words and compare the generated summaries based on accuracy, completeness, and readability.

Python Code:

```
from openai import OpenAI

# Paste your OpenAI API Key here
client = OpenAI(
    api_key="sk-proj-
fXs9TwmdiZQU9td92kDIe9wACeeq8S0wsiGwQRfrbEoJq4VGPLU1Np74KCilODIH8_C
oBWZp8nT3BlbkFJBjvFWrf8sOhvLZ7zO3YVr2MDh2NeIM42jru5-
z2nctrTGHuEgdMAYbFx2lo34b4CY-q8ffMBkA"
)

article = """
Artificial Intelligence (AI) is transforming healthcare by improving disease diagnosis,
enabling personalized treatment, assisting in medical imaging, supporting robotic surgeries,
and enhancing patient monitoring. AI-powered systems help doctors analyze medical data
more
quickly and accurately, leading to better healthcare services and improved patient outcomes.
"""

# ----- Zero-shot Prompt -----
zero_shot = f"""
Summarize the following article in exactly 50 words.

Article:
{article}
"""

response = client.responses.create(
    model="gpt-5.4-mini",
    input=zero_shot
)

print("===== ZERO-SHOT SUMMARY =====")
print(response.output_text)

# ----- One-shot Prompt -----
```

```
one_shot = f"""
Example
```

Article:
Online education allows students to learn from anywhere using digital platforms.

Summary:
Online education enables flexible learning through digital platforms, allowing students to access quality education from anywhere while improving convenience and learning opportunities.

Now summarize the following article in exactly 50 words.

```
Article:
{article}
"""
```

```
response = client.responses.create(
    model="gpt-5.4-mini",
    input=one_shot
)

print("\n===== ONE-SHOT SUMMARY =====")
print(response.output_text)
```

```
# ----- Few-shot Prompt -----
few_shot = f"""
Example 1
```

Article:
Solar energy is a renewable energy source that reduces pollution and electricity costs.

Summary:
Solar energy provides clean, renewable power while reducing pollution, lowering electricity expenses, and supporting sustainable development.

Example 2

Article:
Cloud computing offers on-demand computing resources, scalable storage, and remote access.

Summary:

Cloud computing delivers flexible computing resources, scalable storage, and remote accessibility, improving efficiency while reducing infrastructure costs.

Now summarize the following article in exactly 50 words.

Article:

```
{article}  
"""
```

```
response = client.responses.create(  
    model="gpt-5.4-mini",  
    input=few_shot  
)  
  
print("\n===== FEW-SHOT SUMMARY =====")  
print(response.output_text)
```

Output

===== ZERO-SHOT SUMMARY =====

AI is reshaping healthcare by speeding diagnosis, enabling personalized treatments, improving medical imaging, aiding robotic surgeries, and strengthening patient monitoring. By analyzing medical data faster and more accurately, AI supports doctors in delivering better care, increasing efficiency, and improving patient outcomes across many areas of medicine.

===== ONE-SHOT SUMMARY =====

Artificial Intelligence is transforming healthcare by improving diagnosis, enabling personalized treatment, assisting medical imaging, supporting robotic surgeries, and enhancing patient monitoring. AI-powered systems help doctors analyze medical data faster and more accurately, resulting in better healthcare services, more efficient workflows, and improved patient outcomes across the medical field.

===== FEW-SHOT SUMMARY =====

Artificial Intelligence is transforming healthcare by improving disease diagnosis, enabling personalized treatment, assisting medical imaging, supporting robotic surgeries, and enhancing patient monitoring. AI-powered systems help doctors analyze medical data faster and more accurately, improving healthcare efficiency, service quality, and patient outcomes.

Result

The Zero-shot, One-shot, and Few-shot prompting techniques were successfully implemented using the OpenAI API to summarize an article into 50 words. The generated summaries were compared based on accuracy, completeness, and readability. The Few-shot prompt produced the most accurate, complete, and readable summary among the three prompting techniques.